Newton's second law: how to isolate m and a in $F = m \cdot a$

I want m

1. Write the law. $F = m \cdot a$
2. **a** is **multiplying** **m**: divide **both sides** by a. $\frac{F}{a} = \frac{m \cdot \cancel{a}}{\cancel{a}}$
3. ✓ Result: $m = \frac{F}{a}$

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Shortcut: what is **multiplying** on one side goes to the other side **dividing**.